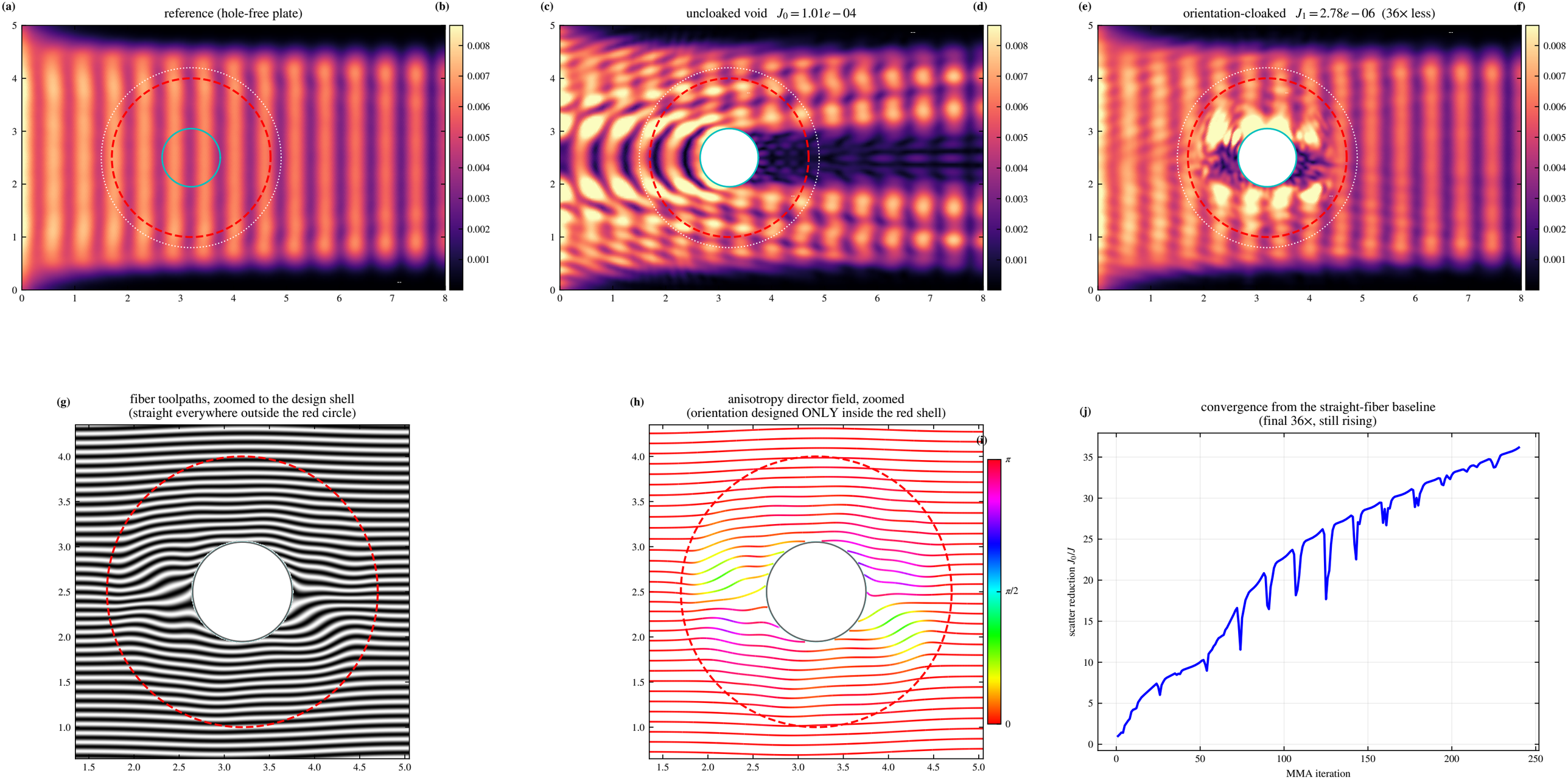

(a)
reference (hole-free plate)
(b)
(c)
uncloaked void $J_0 = 1.01e-04$
(d)
(e)
orientation-cloaked $J_1 = 2.78e-06$ (36× less)
(f)
(g)
fiber toolpaths, zoomed to the design shell
(straight everywhere outside the red circle)
(h)
anisotropy director field, zoomed
(orientation designed ONLY inside the red shell)
(i)
π
$\pi/2$
0
(j)
convergence from the straight-fiber baseline
(final 36×, still rising)
scatter reduction J_0/J
MMA iteration